Singular Value Decomposition

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of singular value decomposition, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

LA-04 / Linear Algebra / Report v1

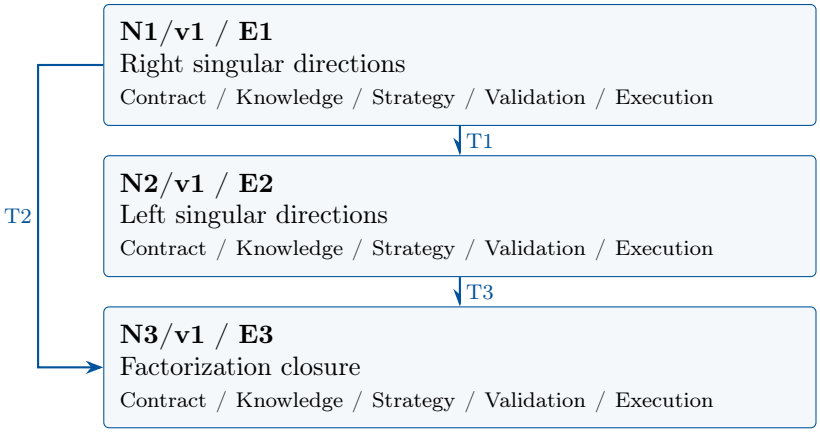
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For every real $m \times n$ matrix A, prove $A = U\Sigma V^T$ with orthogonal U, V and rectangular diagonal Σ having nonnegative singular values.*

A. Foundations and permitted imports

Import LA-03: every real symmetric matrix has an orthonormal eigenbasis. Import orthonormal completion from LA-02 and LA-01's finite-dimensional kernel/rank facts. For $A : R^n \to R^m$, the matrix $A^T A$ is symmetric positive semidefinite. U and V are square orthogonal matrices; Σ is $m \times n$ with nonnegative entries on its rectangular diagonal.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. An orthonormal right basis with $Av_i = 0$ for $i > r$.

Receiving activity. N2: Left singular directions

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Positive σ_i and right eigenvectors; prior exports E1/v1

T2: N1/E1/v1 to N3. An orthonormal right basis with $Av_i = 0$ for $i > r$.

Receiving activity. N3: Factorization closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Both orthonormal bases; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. A full orthonormal left basis whose first r vectors match positive singular directions.

Receiving activity. N3: Factorization closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Both orthonormal bases; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Right singular directions

Object and role. Order positive eigenvalues before zero ones.

Inputs. Spectral theorem on A^TA; foundational imports stated in §1

C — Contract

Own E1/v1: An orthonormal right basis with $Av_i = 0$ for $i > r$.

K — Knowledge

Retain Spectral theorem on A^TA; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Order positive eigenvalues before zero ones.

V — Validation

Negative eigenvalues cannot occur; tiny computed negatives are numerical errors, not theorem data.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Let $v_1, \ldots, v_n$ be an orthonormal eigenbasis of A^TA with eigenvalues λ_i.

N1.2 Since $v_i^T A^T A v_i = \|Av_i\|^2 \geq 0$, all $\lambda_i \geq 0$.

N1.3 Arrange $\lambda_1, \ldots, \lambda_r > 0$ and the rest zero; put $\sigma_i = \sqrt{\lambda_i}$.

N1.4 For $\lambda_i = 0$, $\|Av_i\|^2 = 0$, so $Av_i = 0$.

N1.5 Moreover $\ker(A^TA) = \ker A$ by this same norm identity, so $r = \operatorname{rank} A$.

Export E1/v1. An orthonormal right basis with $Av_i = 0$ for $i > r$.

Acceptance gate. Negative eigenvalues cannot occur; tiny computed negatives are numerical errors, not theorem data. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Negative eigenvalues cannot occur; tiny computed negatives are numerical errors, not theorem data. Recheck dependent nodes N2, N3.

B. N2/v1: Left singular directions

Object and role. Normalize only nonzero images.

Inputs. Positive σ_i and right eigenvectors; prior exports E1/v1

C — Contract

Own E2/v1: A full orthonormal left basis whose first r vectors match positive singular directions.

K — Knowledge

Retain Positive σ_i and right eigenvectors; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Normalize only nonzero images.

V — Validation

The zero singular values require completion, not normalization.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For $i \le r$ define $u_i = Av_i/\sigma_i$.

N2.2 Then $u_i^T u_j = v_i^T A^T A v_j/(\sigma_i\sigma_j) = \lambda_j \delta_{ij}/(\sigma_i\sigma_j) = \delta_{ij}$.

N2.3 Thus these r vectors are orthonormal in R^m, in particular $r \le m$.

N2.4 Complete them to an orthonormal basis $u_1, \ldots, u_m$ without attempting to divide zero images by zero singular values.

Export E2/v1. A full orthonormal left basis whose first r vectors match positive singular directions.

Acceptance gate. The zero singular values require completion, not normalization. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The zero singular values require completion, not normalization. Recheck dependent nodes N3.

C. N3/v1: Factorization closure

Object and role. Check the equality on every basis vector.

Inputs. Both orthonormal bases; prior exports E1/v1, E2/v1

C — Contract

Own E3/v1: The full real rectangular SVD with nonnegative singular values.

K — Knowledge

Retain Both orthonormal bases; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

S — Strategy

Check the equality on every basis vector.

V — Validation

No uniqueness of U or V is asserted.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Let V have columns v_i, U have columns u_i, and Σ have σ_i at (i,i) for $i \leq r$ and zeros elsewhere.

N3.2 For $i \leq r$, $U\Sigma V^T v_i = \sigma_i\, u_i = Av_i$; for $i > r$ both sides vanish.

N3.3 Since v_i form a basis, $A = U\Sigma V^T$.

N3.4 This includes $A = 0$ and rectangular dimensions.

N3.5 The sorted singular values are determined by the eigenvalues of $A^T A$; bases in repeated or zero eigenspaces need not be unique.

Export E3/v1. The full real rectangular SVD with nonnegative singular values.

Acceptance gate. No uniqueness of U or V is asserted. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: No uniqueness of U or V is asserted. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Right singular directions

1. Let $v_1, \ldots, v_n$ be an orthonormal eigenbasis of $A^T A$ with eigenvalues λ_i.
2. Since $v_i^T A^T A v_i = \|Av_i\|^2 \geq 0$, all $\lambda_i \geq 0$.
3. Arrange $\lambda_1, \ldots, \lambda_r > 0$ and the rest zero; put $\sigma_i = \sqrt{\lambda_i}$.
4. For $\lambda_i = 0$, $\|Av_i\|^2 = 0$, so $Av_i = 0$.
5. Moreover $\ker(A^T A) = \ker A$ by this same norm identity, so $r = \operatorname{rank} A$.

B. §4.N2 — Left singular directions

1. For $i \leq r$ define $u_i = Av_i/\sigma_i$.
2. Then $u_i^T u_j = v_i^T A^T A v_j/(\sigma_i \sigma_j) = \lambda_j \delta_{ij}/(\sigma_i \sigma_j) = \delta_{ij}$.
3. Thus these r vectors are orthonormal in R^m, in particular $r \leq m$.
4. Complete them to an orthonormal basis $u_1, \ldots, u_m$ without attempting to divide zero images by zero singular values.

C. §4.N3 — Factorization closure

1. Let V have columns v_i, U have columns u_i, and Σ have σ_i at (i,i) for $i \leq r$ and zeros elsewhere.
2. For $i \leq r$, $U\Sigma V^T v_i = \sigma_i\ u_i = Av_i$; for $i > r$ both sides vanish.
3. Since v_i form a basis, $A = U\Sigma V^T$.
4. This includes $A = 0$ and rectangular dimensions.
5. The sorted singular values are determined by the eigenvalues of $A^T A$; bases in repeated or zero eigenspaces need not be unique.

□

V. WORKED ORCHESTRATION

For $A = [[3, 0], [0, 0], [0, 0]]$, take $V = I_2, U = I_3$ and $\Sigma = A$. The singular values are 3 and 0; only the first left vector comes from normalizing Av_1. The other two are an orthonormal completion.

A. Boundary and hypothesis audit

Trying $u_i = Av_i/\sigma_i$ at $\sigma_i = 0$ is undefined even for the zero matrix. Validation returns that boundary test; Knowledge supplies the kernel identity, and Execution completes an orthonormal basis for the remaining directions.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

Trying $u_i = Av_i/\sigma_i$ at $\sigma_i = 0$ is undefined even for the zero matrix. Validation returns that boundary test; Knowledge supplies the kernel identity, and Execution completes an orthonormal basis for the remaining directions.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.